

Logic

This assignment is the **only** assignment in the course where you do not have to program in Julia/JuMP.

Problem 1

- OR : Define constraints, which relates the variables x and y to v such that if either x or y are 1 then v is forced to be 1.

With Three Constraints:

- Enforce v by x

$$x \leq v$$

- Enforce v by y

$$y \leq v$$

- Enforce x or y by v

$$x + y \geq v$$

With Two Constraints:

- Enforce v by x and y with one constraint:

$$x + y \leq 2 \cdot v$$

- Enforce x or y by v

$$x + y \geq v$$

Problem 2

- AND: Define constraints such that if both x and y are 1 then v is forced to be 1.

- Enforce v by x and y

$$x + y - 1 \leq v$$

- And

$$x \geq v$$

- And

$$y \geq v$$

Problem 3

Problem:

- NOT: Define one constraint such that if x is 1 then v is forced to be 0 and if x is 0 then v is forced to be 1

With One Constraint:

- Enforce v by x

$$v = 1 - x$$